

QTM 347 - Homework 4: Part 3 Problem 1 Report

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Executive Summary

Understanding and measuring intelligence has long been a central focus of education, psychology, and labor economics. Among the most widely used tools in the United States is the Armed Services Vocational Aptitude Battery (ASVAB), a complex assessment originally designed to evaluate aptitudes across science, mathematics, language, and technical spheres. Beyond its military application, as it continues to serve as a required screening test for branches of the U.S. Armed Forces, the ASVAB has become a valuable source of longitudinal data for studying how early cognitive and technical skills relate to later life outcomes.

In this study, we investigated the relationship between measured intelligence and future socioeconomic success using data from the National Longitudinal Survey of Youth (NLSY), with a particular focus on cognitive ability, demographic characteristics, and education. The dataset, IQ.Full.csv, includes individuals whose ASVAB scores were collected in 1981 and whose income was recorded in 2005, alongside demographic variables such as gender, race, family background, and educational attainment. This rich and long-term data allows us to explore questions about predictive modeling of future earnings.

Our analysis centers on three research questions:

1. **Is IQ related to one's successes measured by Income?** - To explore this, we evaluate whether ASVAB-derived intelligence measures are significantly associated with earnings later in adulthood.
2. **Is there evidence to show that Females are under-paid?** - We test for evidence that females may be underpaid relative to males, controlling for ability measures and other demographic factors.
3. **What are the best possible prediction models to predict future income?** - Using regression and decision trees, we identify the most effective models for forecasting future income based on early-life cognitive assessments and demographic features are ensemble methods.

To answer these research questions, we began by applying principal component analysis (PCA) to the ten ASVAB subtests, allowing us to summarize complex cognitive and technical abilities into a smaller number of interpretable components. These components, along with demographic and educational variables, served as the foundation for our regression analyses examining how measured intelligence relates to adult income. We then extended this work by developing a series of decision trees and an ensemble-based predictive model to evaluate how well gender, education, and early-life cognitive measures can forecast later socioeconomic outcomes. The best performance came from a four-model ensemble (fit5), which produced the lowest test error.

Using this final model, we estimated an expected annual income of roughly \$19,250 for a representative individual (“Michelle”), illustrating both the capabilities and limitations of translating cognitive assessments into personalized income predictions.

Exploratory Data Analysis

Our data set IQ.Full.csv is a subset of individuals from the 1979 National Longitudinal Study of Youth (NLSNY79) survey who were re-interviewed in 2006. Data about family composition, personal demographic information such as gender, race, and educational level were collected. Each observation also has a set of ASVAB test scores. We began by loading the dataset and conducting an initial review of its structure, dimensions, and variable types. No missing values were found.

Variable Definitions

Variable	Description
Subject	Individual Subject ID
Race	1 = Hispanic, 2 = Black, 3 = Not Hispanic or Black
Gender	Factor with levels “female” and “male”
Educ	Years of education completed by 2006
Imagazine	A variable taking on the value 1 if anyone in the respondent’s household regularly read magazines in 1979, otherwise 0
Innewspaper	A variable taking on the value 1 if anyone in the respondent’s household regularly read newspapers in 1979, otherwise 0
Illibrary	A variable taking on the value 1 if anyone in the respondent’s household had a library card in 1979, otherwise 0
MotherEd	Mother’s years of education
FatherEd	Father’s years of education
AFQT	Percentile score on the AFQT intelligence test in 1981
Coding	Score on the Coding Speed test in 1981
Auto	Score on the Automotive and Shop test in 1981
Mechanic	Score on the Mechanic test in 1981
Elec	Score on the Electronics Information test in 1981
Science	Score on the General Science test in 1981
Math	Score on the Math test in 1981
Arith	Score on the Arithmetic Reasoning test in 1981
Word	Score on the Word Knowledge Test in 1981
Parag	Score on the Paragraph Comprehension test in 1981
Numer	Score on the Numerical Operations test in 1981

*There are also self-esteem scores that were omitted as predictors

The *Subject* variable was removed, and *Gender* was recoded as a binary factor (0 = female, 1 = male). Because income is strongly right-skewed, we applied a natural log transformation to *Income2005*, creating *log_Income2005* as the primary outcome to stabilize variance and improve the model's fit. Michelle, the final observation in the dataset, was intentionally separated before data splitting so that her income could later be predicted without leakage. The remaining data was partitioned into 70% training, 20% testing, and 10% validation sets.

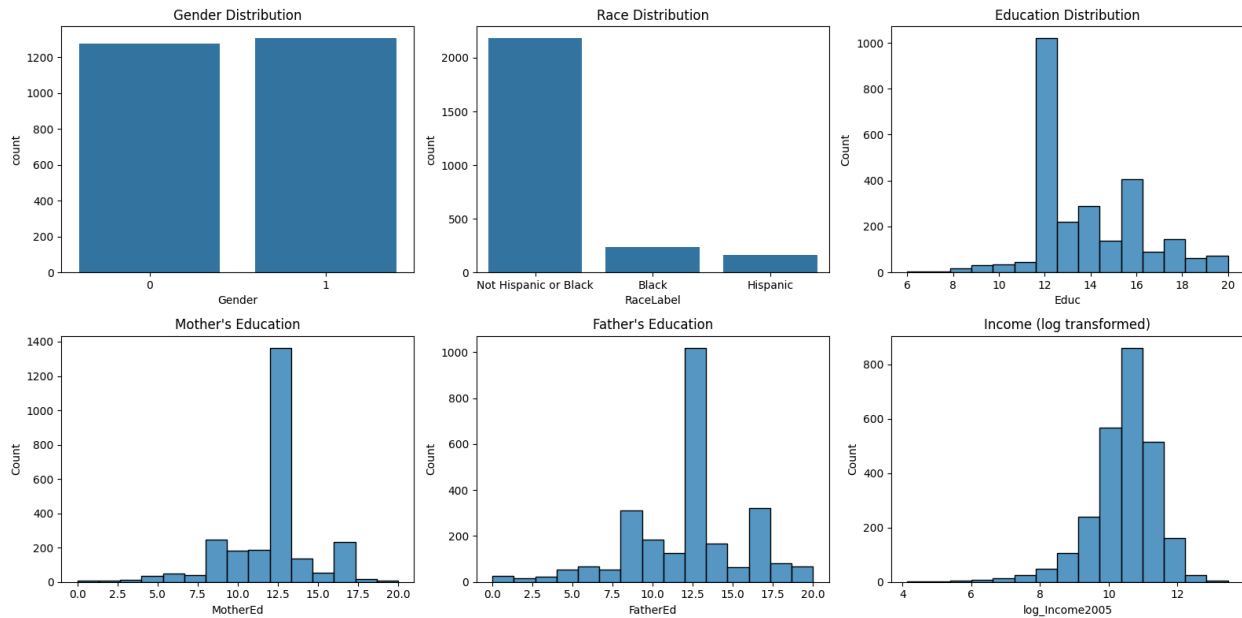


Figure 1: Distribution Plots for Highlighted Variables

To better understand the structure of the dataset, we examined the distributions of key demographic and socioeconomic variables. The gender distribution is nearly balanced, with similar counts of males and females, indicating that comparisons by gender are unlikely to suffer from sample imbalance. The race distribution, however, is skewed, with the majority of respondents classified as Not Hispanic or Black, with substantially smaller groups of Black and Hispanic respondents. This imbalance suggests that racial comparisons will require careful interpretation due to unequal group sizes.

Education-related variables all showed strong central clustering. Years of education showed a pronounced peak at 12 years, reflecting the large proportion of respondents who completed high school, followed by a gradual spread into college-level education. Mother's and father's education demonstrated similar patterns, each with a heavy concentration around 12 years and smaller tails extending toward lower and higher levels of schooling. Finally, the distribution of

the transformed income variable ($\log_Income_{2005}$) showed a much more symmetric and bell-shaped form compared to the raw income (which was strongly right-skewed). The approximately normal distribution of log-income supports the use of linear modeling techniques and confirms that the log transformation successfully reduced skewness and stabilized variance. Together, these exploratory plots reveal a dataset with balanced gender representation, uneven racial composition, concentrated education completion, and a well-behaved outcome variable suitable for regression modeling.

Analyses

Principal Component Analysis

To better understand patterns within the ASVAB cognitive ability measures, we conducted a Principal Component Analysis (PCA) on the ten ASVAB subtests after standardizing each variable to have mean 0 and variance 1. Standardization was essential because the subtests are scored on different scales and would otherwise contribute unequally to the components. The PCA revealed a clear underlying structure in the ASVAB scores. The first principal component (ASVAB_PC1) had strong, positive loadings across all ten subtests, indicating that individuals who performed well on one ASVAB section tended to perform well across the board. This component summarizes overall test performance in a single dimension. ASVAB_PC1 explained approximately 61% of the total variance, reinforcing its role as the dominant source of ability-related differences in the sample.

The second principal component (ASVAB_PC2) reflected specific skill differentiation. Subtests involving numerical reasoning, coding speed, and verbal processing had positive loads, while technical or mechanical knowledge had negative loads. This contrast suggests that ASVAB_PC2 distinguishes individuals with relatively stronger administrative or analytical skills from those who have more technical skills. Unlike PC1, which captures general aptitude, PC2 represents skill specialization. (See Appendix B for specific loadings)

Linear Regression with PCA Components

To evaluate whether cognitive ability influences income, we fit an OLS model predicting log-transformed income from ASVAB_PC1, ASVAB_PC2, gender, and years of education. Both principal components were statistically significant predictors, suggesting that higher general ASVAB performance and stronger administrative or coding skills are associated with higher income. Specifically, a one-unit increase in ASVAB_PC1, a general cognitive ability factor, was associated with approximately a 7% increase in income, while a one-unit increase in ASVAB_PC2, which reflects stronger administrative and numerical processing skills rather than technical skills, corresponded to a 6–7% increase in income. This indicates that both overall cognitive ability and skill specialization have meaningful financial implications.

Gender exhibited a highly significant effect on income. Controlling for ASVAB performance and education, males earned approximately 96% more than females. This gendered difference cannot be attributed to measured ability or schooling. This result provides strong evidence of gender-based income inequality within the sample. Education also behaved as expected, showing a positive and significant association with income. This reinforces the well-established link between higher education and improved labor market outcomes via income.

fit1: Basic Decision Tree Model

The first decision tree model (fit1) was constructed using only gender and education as predictors. This is to provide a baseline comparison for more complex models. Because no tuning parameters or pruning constraints were applied, the tree grew to 30 terminal nodes, most likely overfitting the training data. This overfitting occurs when the model attempts to capture noise or minor fluctuations in the sample, rather than the true underlying patterns, which can reduce generalizability to unseen data. The test MSE for this model was 0.6336, reflecting its limited predictive accuracy despite its complexity.

Despite this, the tree revealed interactions between gender and education, specifically that the effect of education on income differed across gender, as seen in separate splitting structures for men and women. In several branches, women required more years of education than men to reach comparable predicted income levels, reflecting patterns consistent with the gender disparities observed in the regression model. Using fit1 to predict Michelle's income resulted in an estimated annual income of approximately \$22,251.39, which served as a reference point for evaluating how additional predictors and more sophisticated modeling methods improve accuracy and stability in subsequent models.

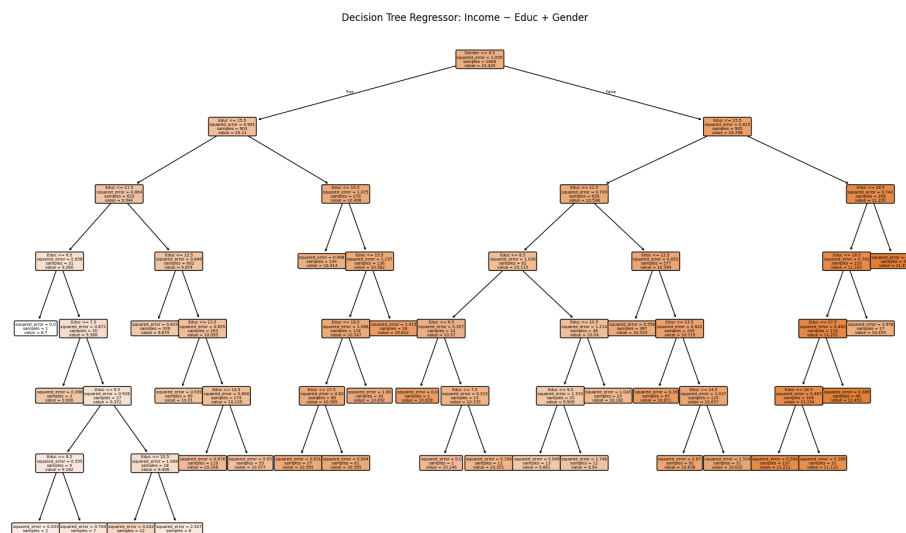


Figure 2: fit1 Decision Tree

fit2: Pruned Tree

To reduce the overfitting observed in the initial decision tree, we built a pruned tree (fit2) using *min_samples_split*=20 and *ccp_alpha*=0.009 to penalize unnecessary splits. This fit2 tree is much simpler, split only once on gender and once on education. This structure reflects a more interpretable model in which gender forms the primary division, followed by an education-based refinement within each gender group. As expected, pruning led to a modest increase in training error relative to the fully grown fit1, since the model no longer attempts to perfectly capture unique observations in the training data. However, the testing error improved slightly, with a test MSE of 0.6271, indicating better generalization and confirming that the complexity of the original tree was not contributing meaningful predictive value. Overall, the pruned model captured the essential relationship between gender, education, and income without introducing unnecessary complexity. Using fit2 to predict Michelle's income, her estimated annual income is approximately \$20,826.89.

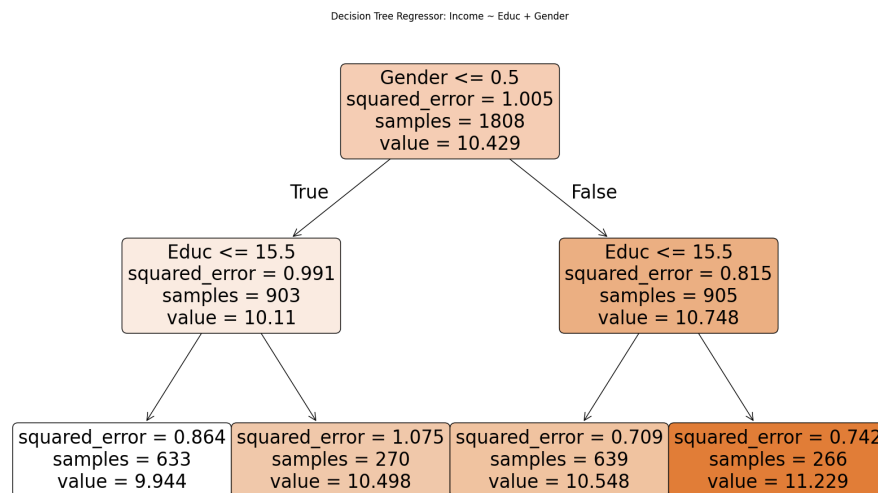


Figure 3: fit2 Decision Tree

fit3: Bagged Trees

To reduce the variance inherent in single decision trees, we implemented a simple bagging approach by constructing two bootstrap samples and fitting a separate decision tree to each. Their predictions were averaged, forming a two-tree bagged ensemble. The motivation for bagging is that, by aggregating multiple unstable models, the ensemble model can smooth out unique fluctuations tied to individual samples and thereby improve predictive accuracy. While bagging theoretically improves predictive accuracy by stabilizing estimates, using only two trees yields limited variance minimization. Bagging is most powerful when a large number of trees, often dozens or hundreds, are combined, allowing random variation across models to cancel out. With just two trees, each model still exerts substantial influence on the final prediction, and the ensemble does not fully capitalize on the stabilizing properties of bagging.

Consistent with this theoretical expectation, the two-tree bagged model performed slightly worse on the test set, with a test MSE of 0.6354, compared to the single-tree models. Using `fit3` to predict Michelle's income, her estimated annual income is approximately \$20,673.42. This result highlights that while bagging is a valuable method for variance reduction, its benefits are best only when a sufficiently large ensemble is used.

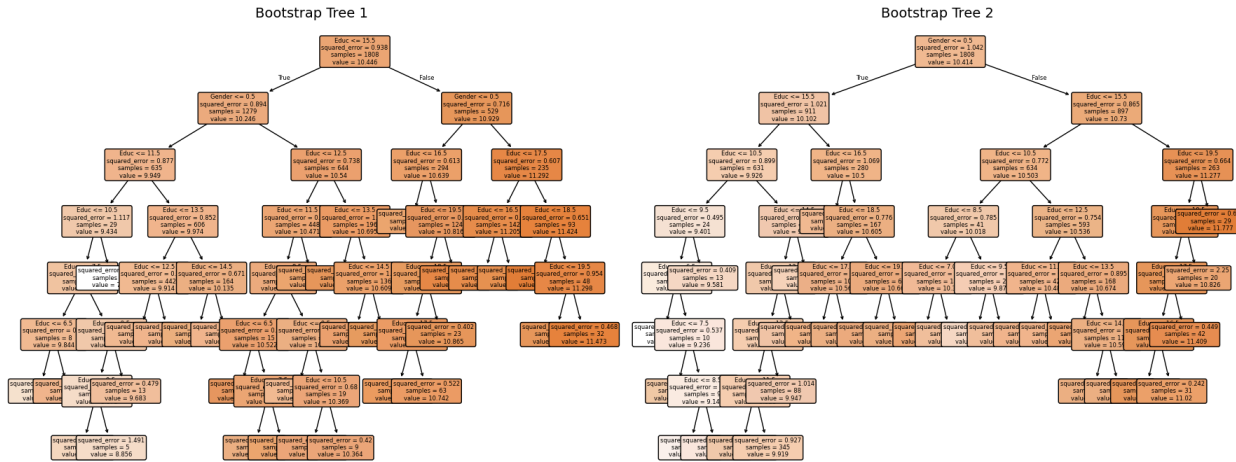


Figure 4: `fit3` Bagged Decision Trees

fit4: Random Forest

We next constructed a random forest model using the full set of predictors, incorporating demographic variables, household characteristics, and the two ASVAB principal components. Raw ASVAB scores and Esteem variables were excluded to reduce dimensionality and to avoid redundancy, ensuring that the forest focused on the most meaningful cognitive measures.

Random forests improve predictive accuracy by aggregating many decision trees, each trained on bootstrap samples and restricted to only consider a subset of predictors at each split. We tuned `n_estimators`, which is the number of trees, by plotting the number of trees against the OOB (out-of-bag) error to see that the OOB (out-of-bag) MSE decreased sharply over the first 25 trees and then stabilized. Beyond approximately 25 trees, the test MSE remained virtually unchanged, indicating that additional trees did not improve performance of the model. This behavior is consistent with the theoretical framework that random forests converge as the number of trees grows, but meaningful gains typically occur early in the process. We can see that initially, increasing numbers of trees is better in minimizing and reducing error. We also tuned `max_features`, the number of features considered when looking for the best split at each node in a tree. This hyperparameter controls the level of decorrelation among trees, where smaller values increase diversity but may weaken individual tree strength, while large values risk trees becoming too similar. We selected the size yielding the lowest OOB error, where the optimal value was `max_features=2`.

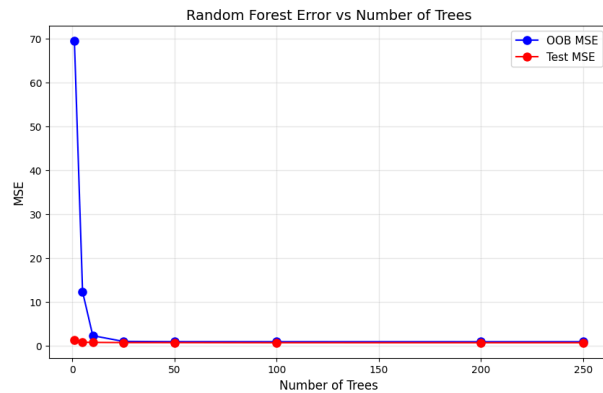


Figure 5: $n_estimators$ Tuning Graph

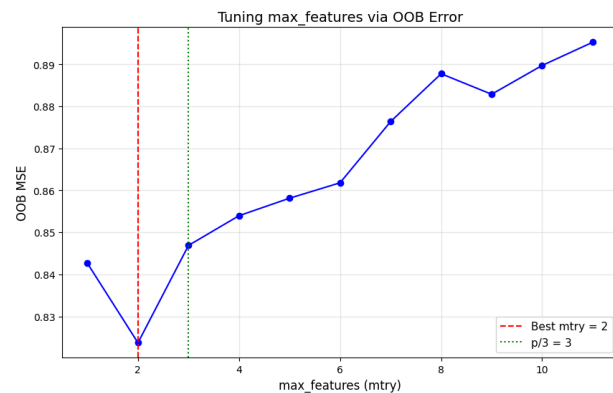


Figure 6: $max_features$ Tuning Graph

The final tuned model (fit4) achieved a test MSE of 0.6326, which is less than the unpruned tree and the bagged model, but greater than the pruned decision tree. Although the OOB error slightly overestimated the true test error, it correctly identified the optimal hyperparameters ($n_estimators$ and $max_features$). The random forest produced smoother and more reliable predictions, reducing overfitting relative to the earlier models. Michelle's predicted income under the fit4 model was approximately \$14,331, reflecting the model's tendency to shrink extreme predictions toward the center due to averaging across many trees.

fit5: Bagged fit1 - fit4 Trees

To combine the strengths of all previous models, we constructed an ensemble/bagged tree model by averaging predictions from fit1 (unpruned tree), fit2 (pruned tree), fit3 (bagged trees), and fit4 (random forest). This approach leverages the principle that aggregating diverse models can produce more stable and accurate predictions than any single model alone. This model (fit5) achieved the lowest test MSE of approximately 0.609, outperforming all other models. When evaluated on the validation set, the ensemble achieved a validation MSE of 0.7362, which is to be expected due to additional out-of-sample uncertainty. Although higher than the test MSE, this still represents reasonably strong performance and indicates that the ensemble generalizes well beyond the training environment. It also performed reasonably well on the validation set, indicating strong out-of-sample stability. By blending simple models, complex models, and bagged models, fit5 reduced variance while preserving the important structure of the data. Using the final ensemble, we predicted Michelle's income to be approximately \$19,250, which is a fairly balanced estimate that incorporates multiple modeling perspectives.

Model	Test MSE
fit1	0.6336
fit2	0.6271
fit3	0.6354

fit4	0.6326
fit5	0.6093

Table 1: Test MSE Values for all Decision Tree Models

Limitations

While the models developed in this analysis provide meaningful insight into the predictors of income and offer reasonably strong performance relative to one another, several limitations should be acknowledged. First, the mean-squared errors (MSEs) for all models were relatively high, even on the log-income scale. This indicates that a significant amount of variability in income remains unexplained by the available predictors and suggests that income is influenced by many unobserved factors not captured in the dataset. Given the complexity of real-world income determination, which is influenced by factors like economic conditions, labor market dynamics, social networks, career choices and personal circumstances, this level of unexplained variance is not surprising.

Additionally, the final bagged model (fit5), although the best performing model on the test set, showed a higher MSE when evaluated on the independent validation set. This increase highlights the inherent uncertainty in out-of-sample prediction and dictates that the model may still exhibit some degree of overfitting or sensitivity to sample-specific patterns. To combat this, a more robust assessment could be obtained with larger samples or alternative modeling approaches. Another limitation with this study is that the dataset is observational and subject to omitted variable bias. Because the dataset is not experimental, all estimated relationships must be interpreted as associations rather than causal effects. Additionally, key determinants of income, such as occupation, industry, family structure, etc. were not available. The larger coefficient on gender, while statistically significant, may also partly reflect structural or societal factors not modeled explicitly.

Conclusion

Through the combined use of PCA, linear regression, and several tree-based models, this study identified a set of consistent and meaningful predictors of income. Cognitive ability, as summarized by the principal components derived from the ASVAB, was a statistically significant and important predictor even after accounting for gender and years of education. Both ASVAB_PC1, representing overall cognitive aptitude, and ASVAB_PC2, reflecting skill specialization, contributed positively to income, suggesting that both general intelligence and specific cognitive strengths play a role in shaping long-term economic outcomes.

Gender disparities emerged as one of the most striking findings. Across all models, males earned substantially more than females, even when holding education and cognitive ability constant.

This pattern aligns with broader labor market trends and underscores the presence of structural or unmeasured factors contributing to gender-based income inequality. Education also exhibited a robust, positive relationship with income, reaffirming its well-established role as a determinant of socioeconomic status.

The tree-based models provided additional insights by uncovering interaction effects and offering interpretable structures that illuminate how variables combine to influence income. However, while individual trees were intuitive, their predictive accuracy varied due to their inherent instability. Bagged/ensemble methods helped address this limitation. By averaging predictions from simple trees, pruned trees, bagged trees, and random forests, the final bagged model (fit5) achieved the lowest test MSE and demonstrated strong stability on the validation set. This improvement illustrates the benefits of combining models with different strengths, reducing variance, and avoiding overfitting. For these reasons, we recommend fit5 as the final predictive model. Its balanced design integrates interpretability, flexibility, and robustness, making it well-suited for forecasting income within this dataset. Using the ensemble, Michelle's predicted income was approximately \$19,250, demonstrating how such models can provide individualized predictions grounded in multiple analytical perspectives. Overall, this analysis highlights the value of combining statistical and machine learning approaches to better understand income determinants and to generate more reliable predictions. While limitations remain, the methods employed here offer a strong foundation for future work and provide meaningful insight into how cognitive ability, education, and demographic factors shape economic outcomes such as income.

Appendix A - Data Cleaning & Preprocessing Details

```
#Dropping first column
```

```
iq = iq_data.drop('Subject', axis=1)
```

```
iq.head()
```

```
#Setting categorical variables as features
```

```
#Female = 0 ; Male = 1
```

```
iq['Gender'] = pd.Categorical(iq['Gender'])
```

```
iq['Gender'] = iq['Gender'].cat.codes
```

```
iq.head()
```

```
#Log transformation of Income
```

```
iq["log_Income2005"] = np.log(iq["Income2005"])
```

```
iq = iq.drop('Income2005', axis=1)
```

```
iq
```

```
#Pull out last row
```

```
michelle = iq.iloc[-1:]
```

```
display(michelle)
```

```
new_iq = iq.iloc[:-1]
```

```
#Splitting data
```

```
from sklearn.model_selection import train_test_split
```

```
#70% train, 30% temp (test+validation)
```

```
train_iq, temp_iq = train_test_split(new_iq, test_size=0.30,  
random_state=0)
```

```
#Split temp into 20% test and 10% validation
```

```
test_iq, val_iq = train_test_split(temp_iq, test_size=(1/3),  
random_state=0)
```

Appendix B - PC1 and PC2 Loadings of ASVAB

Loadings for PC1 and PC2 of ASVAB:

	PC1	PC2
Science	0.351853	-0.141819
Arith	0.354309	0.048735
Word	0.349606	0.061478
Parag	0.326264	0.189665
Numer	0.274897	0.451772
Coding	0.233865	0.514659
Auto	0.272229	-0.473364
Math	0.335974	0.142564
Mechanic	0.316066	-0.334696
Elec	0.323800	-0.335213

Table 1: Loadings for PC1 and PC2

Appendix C - Linear Regression Results

Linear Regression Results:

=====

Intercept (β_0): 10.4376

Slope (β_1): 0.1317

Equation: win = 10.438 + 0.132 × payroll

Model Performance:

R-squared: 0.112

Root Mean Squared Error: 0.9178

Output 1: log_Income2005 ~ ASVAB_PC1

Linear Regression Results:

=====

Intercept (β_0): 10.4376

Slope (β_1): -0.0753

Equation: win = 10.438 + -0.075 × payroll

Model Performance:

R-squared: 0.009

Root Mean Squared Error: 0.9697

Output 2: log_Income2005 ~ ASVAB_PC2

OLS Regression Results

=====

Dep. Variable:	log_Income2005	R-squared:	0.217
Model:	OLS	Adj. R-squared:	0.215
Method:	Least Squares	F-statistic:	178.2
Date:	Tue, 09 Dec 2025	Prob (F-statistic):	6.05e-135
Time:	15:22:20	Log-Likelihood:	-3281.4
No. Observations:	2583	AIC:	6573.
Df Residuals:	2578	BIC:	6602.
Df Model:	4		
Covariance Type:	nonrobust		

=====

	coef	std err	t	P> t	[0.025	0.975]
-----	-----	-----	-----	-----	-----	-----
const	9.0801	0.118	76.681	0.000	8.848	9.312
ASVAB_PC1	0.0695	0.008	8.321	0.000	0.053	0.086
ASVAB_PC2	0.0630	0.018	3.441	0.001	0.027	0.099
Gender	0.6706	0.043	15.431	0.000	0.585	0.756
Educ	0.0733	0.008	8.844	0.000	0.057	0.090

=====

Omnibus:	1076.427	Durbin-Watson:	1.946
Prob(Omnibus):	0.000	Jarque-Bera (JB):	7286.513
Skew:	-1.832	Prob(JB):	0.00
Kurtosis:	10.368	Cond. No.	99.3

=====

===

Output 3: OLS Regression Results of $\log_Income2005 \sim ASVAB_PC1 + ASVAB_PC2 + Gender + Educ$

Appendix D - Michelle's Predicted Incomes

Model	Log Income Prediction	Income Prediction (\$)
fit1	10.010	\$22,251.39
fit2	9.944	\$20,826.89
fit3	9.937	\$20,673.42
fit4	9.570	\$14,331.49
fit5	9.865	\$19,249.82
MICHELLE TRUE VALUE	10.714	\$45,000